

Zhiheng Zhang

m15601654187@163.com | zhihezhang@student.unimelb.edu.au
github.com/zhiheng-zhang-Mera

Melbourne, Australia

Research Profile

Computer Science researcher and research engineer focused on **secure and reliable autonomous coding agents, program-analysis-informed validation, and capability-bounded tool use**. I am interested in connecting long-horizon agent runtimes with software-security and testing signals so that autonomous actions remain inspectable, constrained, and verifiable.

Education

University of Melbourne

Master's degree in Computer Science

Melbourne, Australia

Expected 2026

University of British Columbia

Bachelor of Science in Computer Science

Vancouver, Canada

Completed Summer 2024

Selected Research and Engineering Projects

DS-Hns — *Long-horizon autonomous software-work runtime*

2026–present

github.com/zhiheng-zhang-Mera/DS-Hns

- Building an execution layer for unattended software work around the official DeepSeek Harness interface, with optional capabilities isolated as extensions.
- Implemented ordered/scheduled task queues, hardware-adaptive local concurrency, unified task lifecycle and terminal events, task history, and terminal notifications.
- Designed extension-failure isolation, restart/recovery-oriented execution, and an idempotent Windows installer with explicit optional-plugin flow.
- Using the system as a testbed for **reliable coding agents, persistent execution, recovery, scheduling, and completion verification**.

Codex Boss — *Multi-agent orchestration and evidence-based acceptance platform*

2026–present

github.com/zhiheng-zhang-Mera/Codex-Boss

- Building a platform for long-running AI workflows in which multiple agents can plan, execute, critique, and adjudicate work while preserving durable task state.
- Developed platform foundations spanning durable state/events, capability boundaries, plugin isolation, knowledge/data lifecycle, dogfooding, and acceptance/verification.
- Investigating how autonomous systems can distinguish genuine task completion from superficially plausible but weak evidence.
- Current semantic-acceptance work deliberately retains unresolved failure cases: a vacuous case is rejected, while a meaningful case still exposes a false negative in the evidence reader rather than being reported as solved.
- Current regression evidence on the relevant branch includes **2,508 unit tests across 215 files**, plus Electron, renderer, and test typechecking.

Privacy Lens — *Trustworthy-software / privacy project*

- Explored privacy-risk interpretation and user-facing warning mechanisms with attention to bounded interpretation, auditability, evidence boundaries, and trustworthy software behavior.
- Supports broader interests in trustworthy AI, software governance, and high-stakes automated systems.

Quant-Ultra — *Empirical quantitative-research and decision-system project*

2026–present

github.com/zhiheng-zhang-Mera/Quant-ultra

- Developed a personal quantitative-research platform with emphasis on robustness, temporal evaluation, data integrity, risk-aware decision making, and reproducible workflows.
- Shifted the project away from headline-return claims toward evaluation quality, non-stationary data, temporal leakage, walk-forward reasoning, and auditable experiment infrastructure.

Research Interests

Software security and reliability; program analysis and fuzzing; security of tool-using agents; capability and permission boundaries; coding-agent validation; recovery from unsafe or partial execution.

Technical and Research Strengths

- End-to-end research engineering: problem formulation, system design, iterative implementation, testing, debugging, and empirical evaluation.

- Multi-module software design with explicit persistence, lifecycle, extensibility, isolation, recovery, and acceptance criteria.
- Git/GitHub-based experimental development, branch-based iteration, regression testing, and reproducible project documentation.
- Practical multi-AI workflow design: using multiple AI systems as coding/research collaborators while independently reviewing plans, evidence, failure modes, and completion claims.

Current Research Direction

I want to investigate **secure and verifiably reliable tool-using coding agents**: combining runtime traces, capability boundaries, program-analysis/testing signals, and explicit acceptance criteria to detect unsafe actions, validate generated changes, and recover safely when evidence remains incomplete.